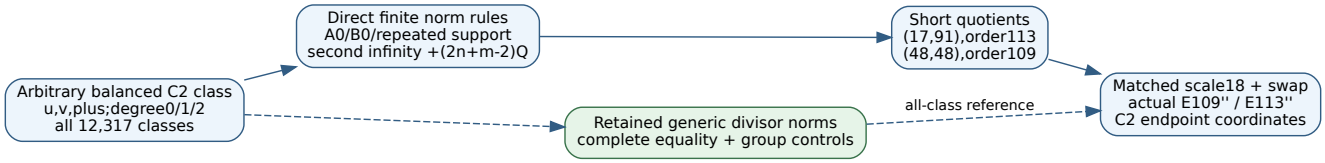
Direct C2 norms for the successor transfers

2026-11-12 · CT25 · exact F101 model · scientific standing unsigned

Both C2 elliptic norms now use direct base-field formulas on arbitrary balanced classes. Every one of the 12,317 rational classes agrees with the retained generic divisor norms, including repeated support and both balanced infinities. The production and generic Phi2/Psi2 graphs also agree on every input. This integrates the same mathematical maps with an explicit C2 factory. No measured speedup is claimed. PCC0355ab/d14c30/158476. [pcc:unverified]



Exact model and matching

C2 is $y^2=x^6+22x^4+10x^2+19$. The covers are(x^2,y) and $(19/x^2,19y/x^3)$. Their long cubics are[19,10,22,1] and[58,14,10,1]. Short shifts41/37 give curves(17,91)/(48,48), with orders113/109. Scale18 identifies them with actual targets(23,7)/(59,27), respectively; the chart swap orders the actual array E109‘‘,E113‘‘. The generic reciprocal chart uses25/x and$19y/x^3$, so its ordinate and sign are unchanged. [pcc:unverified]

The old C0/C1 constructor keeps its whitelist and still rejects C2. A separate explicit factory supplies[19,10,22] to the existing formula engine. Balanced validation, field, polynomial, elliptic and group engines remain in place. No other field, curve family or catalog is accepted. [pcc:unverified]

Complete exceptional-case coverage

The disjoint degree/zero categories contain [3,4,232,2,216,232,11628] classes. The comparison includes118 repeated supports and238 nonzero infinity corrections. Degree0/1/2, A0/B0, tangent/opposite cases and source x0 all have explicit formulas. Every output is checked on its actual quotient; all paired additions, orders, identity and wrong-model controls pass. Source01/receipt0355ab are frozen. [pcc:unverified]

Proof, references and final acceptance

Why these are the generic Picard norms

For $u=x^2+A\,x+B$ and $v=V0+V1\,x$, A nonzero gives a symmetric addition slope -V1/A on the first quotient. The ordinate is eliminated using the support relation, so individual support roots are not selected. Valid repeated support carries its Mumford ordinate jet. At A0, parity forcesV0V1=0: coincident images are doubled and opposite images sum to the origin. [pcc:unverified]

For the second quotient with B nonzero, invert the support algebra and reduce19$(V0t^3+V1t^2)$ modulo $t^2+(A/B)t+1/B$. At B0, the x0 point maps to the origin and the remaining point is evaluated explicitly. Both source infinities map to the first origin and to±Q on the second quotient, Q=(37,19). Thus the second correction is(2n+m-2)Q. Finite-cover pushforward and its projective completion identify the formulas with the generic Picard norms, including multiplicities. Finite output equality is not used as a density argument. [pcc:unverified]

Read SUCCESSOR_NORMS.md/PCC7fb16a, identical to upstream95693e, and the frozen generic DIRECT_NORMS proof. Seven SuccessorNorms Lean lemmas check exact C2 curve, reciprocal inverse, shift and scale identities, PCCc2aaaa. The finite fields, balanced Picard objects and elliptic functoriality remain written bridges. Formalization is partial. [pcc:unverified]

References remain genuine

The original C2 coordinate evaluator remains generic. Generic Phi2 and quarter-Psi2 versions select old divisor norms; the naive halving route keeps generic Phi2 evaluation. Complete Phi2/Psi2 tests compare production and generic outputs on both12,317-class domains and retain dual7, bijections, paired sums, descent and wrong-model rejection. Both original 109/113 source components retain naive comparisons, reverse196 with inverses104/64, actual endpoint4 beta alpha and geometric controls. [pcc:unverified]

Complete changed guard and scope

Final check.sh04 passes25 native/81 reference unit plus five integration tests, one pre-existing ignored benchmark, all source guards, formatting and strictClippy. Three final arithmetic hashes match frozen source04. Complete96,421-byte raw output is standard-xz archived11,612bytes, with raw/archive SHA256 and exact roundtrip, PCC03b76b. Source/receipts and negative history are retained without changing old claims. [pcc:unverified]

The constructor/endpoint change is implemented and locally checked; new exact-head CI and stacked merge requirements remain separate. Prior CT24 head43a CI43125719626 succeeds25/80+five, gate passes/sign skips, PCCd942e6. All scientific standing remains unsigned. [pcc:unverified]

No stage, pipeline or ECDLP algorithm improvement is measured here. Matched comparisons with the index-calculus baseline and rho remain owed under an admitted protocol. S3/livebudget and vault/ledger limits persist. Next CT26 tests deferred construction of the quadratic table used by the reference evaluator; the proposed lower-degree rewrite was a recorded no-op because its existing formulas are already algebraic. [pcc:unverified]